Library

Techmapping chooses library cells that cover AIG cuts

The idea

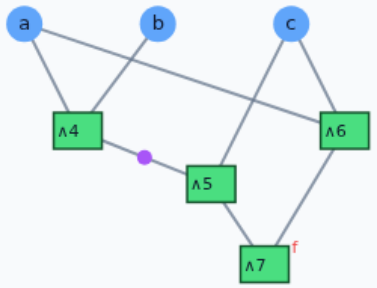
- Six cells: area and delay costs; INV for complemented edges

Six cells

LIBRARY

Six cells

Step 1 / 5 · six · module02-07-library-cells



Explanation

Library size is six.

- INV…AOI21

```
lib=6
```

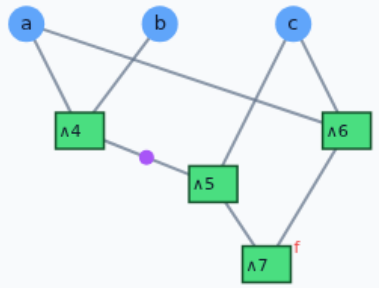
Six cells

AND2

LIBRARY

AND2

Step 2 / 5 · and2 · module02-07-library-cells



Explanation

AND2 area=2 delay=2.

- Default map

```
area 2
```

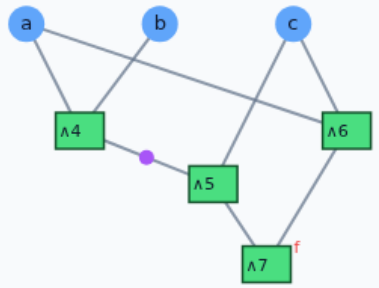
AND2

NOR2

LIBRARY

NOR2

Step 3 / 5 · nor2 · module02-07-library-cells



Explanation

NOR2 covers AND(~,~).

- Both comp

```
Mapping hint
```

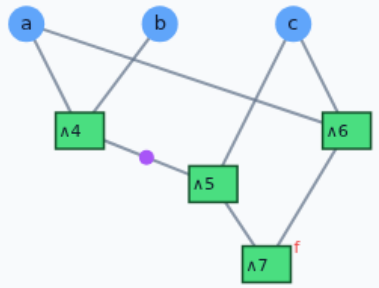
NOR2

INV

LIBRARY

INV

Step 4 / 5 · inv · module02-07-library-cells



Explanation

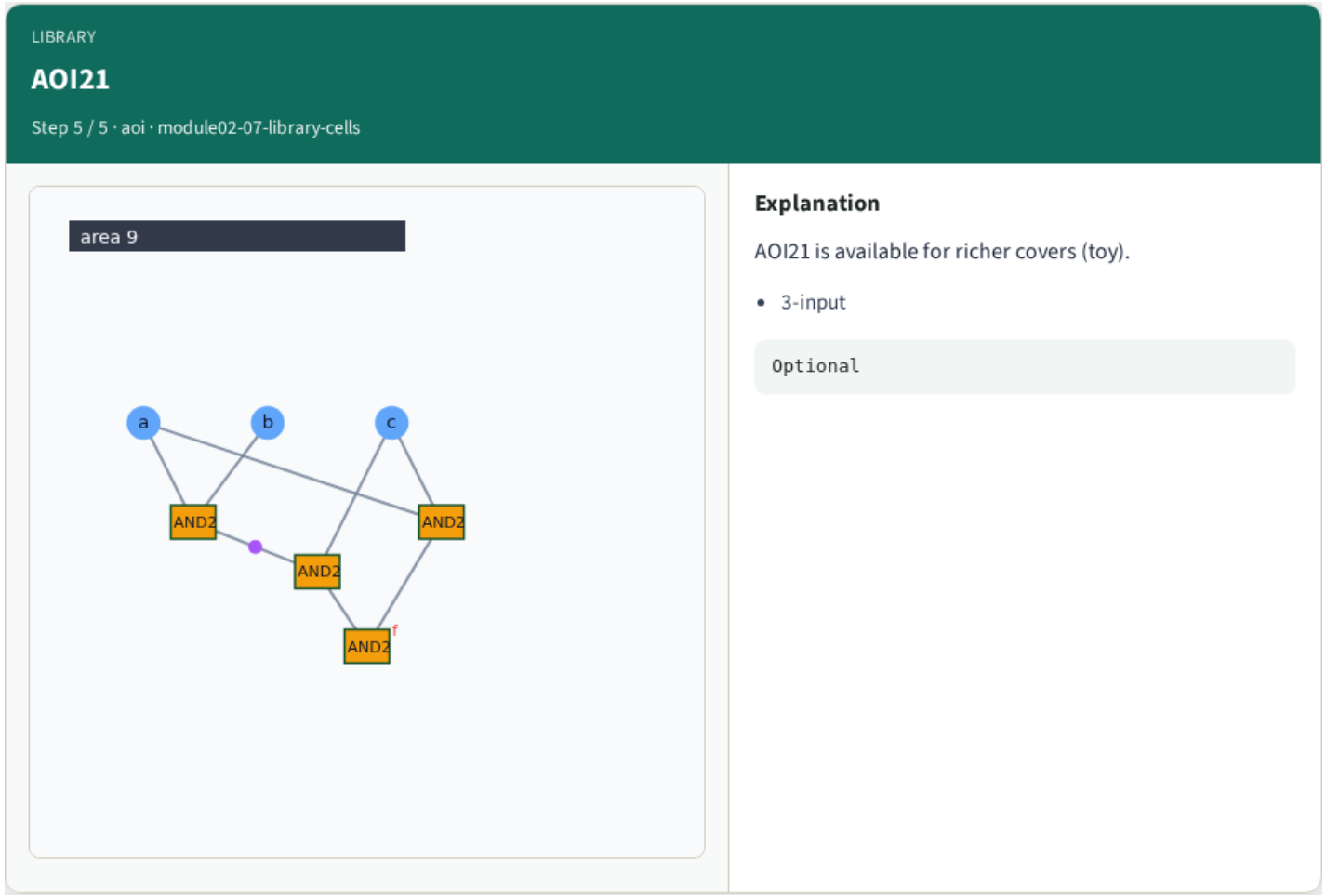
INV area=1 for edge bubbles.

- Complement cost

```
area 1
```

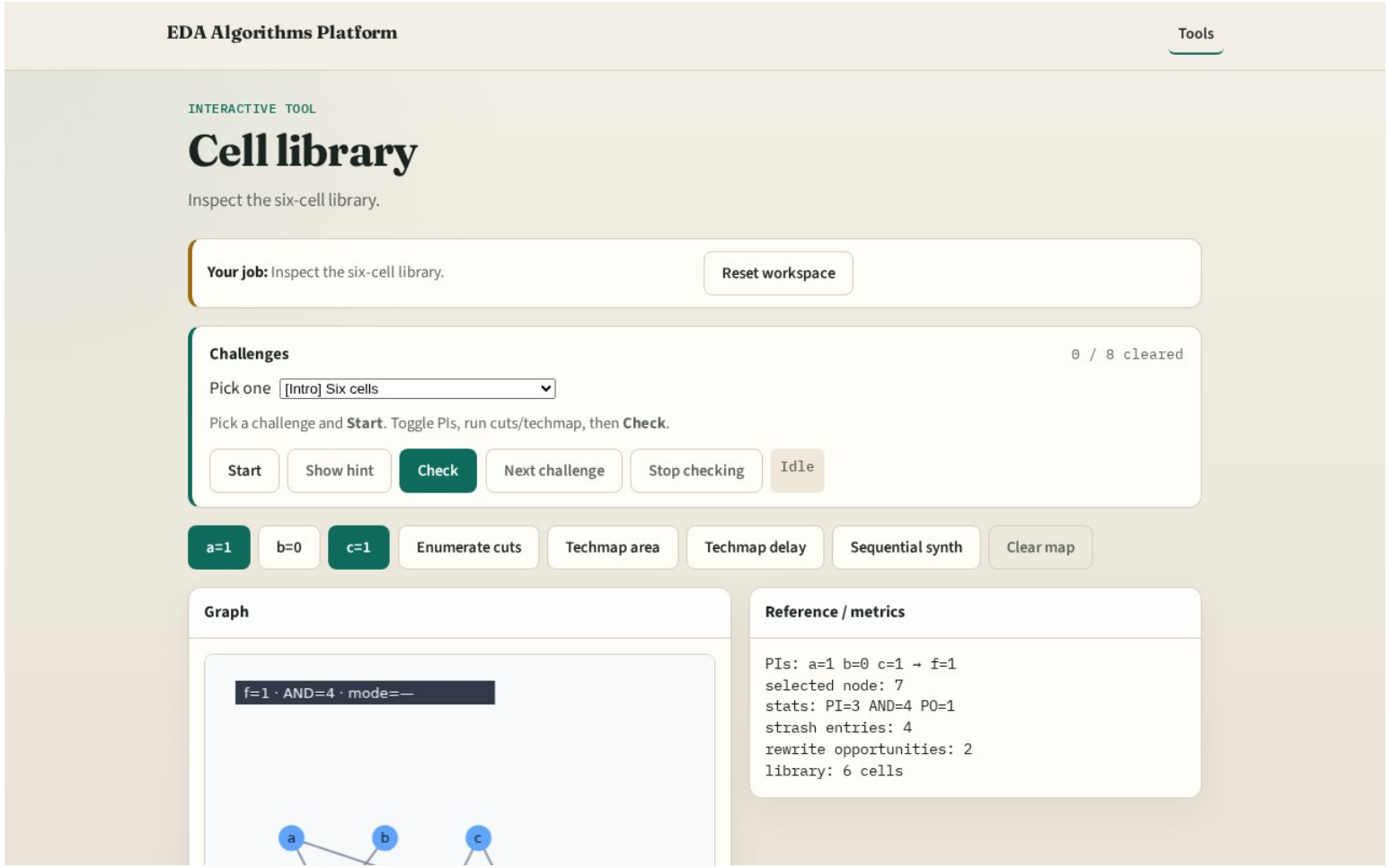
INV

AOI21


LIBRARY
AOI21
Step 5 / 5 · aoi · module02-07-library-cells
area 9
a
b
c
AND2
AND2
AND2
AND2
f
Explanation
AOI21 is available for richer covers (toy).
3-input
Optional

AOI21

Browser lab track



Browser lab starter

Implement track

- Print data['library']; assert len==6

Pitfalls

- Ignoring area when comparing
- Mapping without INV cost

Your turn

- Next: area-oriented techmap cover